Computational Chemistry Laboratory

Vibrational Spectroscopy of the $(H_2O)_{11}$ Cluster

Provided Laboratory Files

Initial coordinates of the 11-water cluster

Reference *ab initio* Cartesian vibrational spectrum

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1 Introduction and Objectives

Vibrational spectroscopy is a fundamental computational tool for probing molecular structures, dynamics, and intermolecular interactions, such as the complex, highly fluxional hydrogen-bonding network present in water clusters. In this laboratory, we will explore the vibrational properties of an 11-molecule water cluster, $(H_2O)_{11}$. We will utilize the semi-empirical extended tight-binding method (**GFN2-xTB**) coupled with the **ORCA** quantum chemistry package. To orchestrate the workflow and extract the spectra, we will employ the custom `dragonball-vispec` package. We will compare two computational approaches:

1. **Static (Harmonic) Approach**: Computing the Hessian matrix at the 0 K optimized geometry to extract harmonic frequencies and normal modes.
2. **Dynamic (Anharmonic) Approach**: Running a Born-Oppenheimer Molecular Dynamics (BOMD) simulation in the NVE ensemble and computing the spectrum directly from the Cartesian trajectory.

Finally, we will benchmark our semi-empirical xTB results against a provided full *ab initio* Cartesian vibrational spectrum.

2 Step 1: Geometry Optimization

To obtain a physically meaningful static spectrum, the cluster must first be relaxed to a local minimum on the Potential Energy Surface (PES). If a frequency calculation is performed on an unoptimized structure, imaginary frequencies will appear, indicating the atomic forces are not zero.

3 Step 2: Hessian and Harmonic Frequencies

Next, we calculate the matrix of second energy derivatives (the Hessian) with respect to nuclear coordinates at the optimized geometry. This confirms we have found a true minimum and yields the harmonic vibrational spectrum.

4 Step 3: Molecular Dynamics Simulation

The harmonic approximation neglects thermal effects and anharmonicity. To incorporate these, we run a short Molecular Dynamics (MD) using quantum harmonic initial conditions. To correctly integrate the fast O–H stretching motions (which vibrate around 3500 cm^{-1}), a very small time step is required.

5 Step 4: Computing the Dynamic FT Cartesian Spectrum

With the MD trajectory generated, we use `dragonball-vispec` to calculate the Velocity Autocorrelation Function (VACF) from the Cartesian coordinates and apply a Fourier Transform to obtain the Cartesian vibrational spectrum (also known as the Vibrational

Density of States). We will also see the differences with respect to the normal mode-wise decomposition of the vibrational spectra.

6 Step 5: Comparison and Lab Report Analysis

You now possess three sets of data:

- The xTB Static Harmonic Spectrum (`xtb_harmonic.dat`).
- The xTB Dynamic Spectrum from MD (`xtb_dynamic.dat`).
- The full *ab initio* Cartesian vibrational spectrum (`abinitio_reference.dat`).

6.1 Plotting the Spectra

Using a plotting software of your choice (e.g., Python with Matplotlib, Origin, or Excel, Flying Numbus GUI or XMgrace), overlay the three spectra on a single graph (Wavenumber vs. Intensity). Normalizing the intensities to a maximum of 1.0 will greatly aid in the visual comparison of peak shapes.

6.2 Questions for the Final Report

Include your plot in your lab report and address the following questions:

1. **Harmonic vs. Anharmonic:** Examine the O–H stretching region ($\approx 3200 - 3600$ cm^{-1}). Is there a noticeable frequency shift (red shift or blue shift) between the static harmonic xTB modes and the dynamic MD modes?
2. How well does the xTB dynamic spectrum reproduce the main features of the high-level *ab initio* reference? Discuss the trade-off between computational cost and accuracy.
3. Identify the three main band regions of the water cluster (Librationem modes, H–O–H Bending, and O–H Stretching). Which modes seem to be most sensitive to the dynamic breaking and forming of hydrogen bonds?